Amedeo Carraro

AI Engineer & Software Architect

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I am an AI Engineer and Software Architect: I design and build LLM systems end to end, from retrieval and prompts to evaluation and deployment, on self-hosted and cloud models: RAG, tool-calling agents and real-time voice agents. MSc in Computer Engineering (Artificial Intelligence and Robotics), with a background in deep learning for computer vision and time series.

PROFESSIONAL EXPERIENCE

AI Engineer, Terya (TeamSystem Group), Bassano del Grappa, Italy — Nov 2025 – Present

- **Chatbot on hundreds of technical manuals.** Answers service-desk questions over 400+ product manuals with step-by-step replies that cite their sources and show the manuals' original images. Hybrid dense + full-text retrieval with rank fusion and cross-encoder reranking, checked by a test harness that replays real support cases. *FastAPI, PostgreSQL/pgvector*
- **Speech-to-speech voice agent for support calls.** Answers calls in real time with barge-in, identifies the caller's store despite transcription errors and opens the help-desk ticket through function calling. *Python asyncio, OpenAI Realtime API*
- **Code knowledge agent.** Turns the internal codebase into documentation that can be queried in plain language, served as context through an 11-tool MCP server: faster development, fewer tokens on paid models. *.NET, PostgreSQL/pgvector, K3s*
- **Tool-calling orchestration for the ERP.** Tools discovered at runtime, JSON Schema-validated arguments with model self-correction, confirmation gates on destructive actions and a per-call audit trail. *FastAPI, SSE streaming*
- **Self-hosted inference.** Open-weight LLM with embedding and reranking models on an on-premises NVIDIA DGX Spark serves all the systems listed here, with OpenAI as fallback and usage tracking for every request. *vLLM, Ollama, Qwen*

Research Intern, Deep Learning (MSc thesis), University of Padua, Italy — Sep 2024 – Mar 2025

- Variable-star classification from Zwicky Transient Facility (ZTF) light curves in PyTorch, with 5-fold cross-validation: 2D-CNN at 88.7% accuracy on 10 classes and 99.95% on 5; hybrid CNN-LSTM at 90.0%, training about 3 times faster.

Full Stack Developer Intern, ITReview (TeamSystem Group), Padua, Italy — Nov 2023 – Dec 2023

- Web applications with React, Node.js/Express and REST APIs on PostgreSQL and MongoDB; database response times cut by about 15% through indexing and query refactoring, measured with EXPLAIN ANALYZE.

Deep Learning & DevOps Intern, AppLibra S.r.l., Thiene, Italy — Oct 2021 – Dec 2021

- Real-time computer-vision check that detects bottle-cap defects on a production conveyor belt: YOLOv5 (PyTorch) and OpenCV, from dataset annotation to GPU training and Docker-based inference on Linux.
- Private NuGet server (BaGet in Docker) for shared .NET libraries; ERP customisations for Microsoft Dynamics NAV.

SELECTED PROJECTS

- **LLM fine-tuning with QLoRA** (group project, University of Cyprus, 2024). Mistral-7B in 4-bit, adapted with LoRA (Hugging Face PEFT) on a single GPU; BLEU 62%, ROUGE-1 81%, METEOR 78%.
- **PCB defect detection** (personal, 2025). YOLOv8 trained on DeepPCB for six defect types, served with FastAPI and packaged with Docker.

EDUCATION

MSc in Computer Engineering: Artificial Intelligence and Robotics, University of Padua — Sep 2022 – Mar 2025
Erasmus+ semester, MSc courses in Artificial Intelligence, University of Cyprus — Jan 2024 – Jun 2024
BSc in Computer Engineering, University of Padua — Sep 2016 – Mar 2022

TECHNICAL SKILLS

LLMs and agents: AI agents, tool/function calling, Model Context Protocol (MCP), structured outputs, streaming, RAG, hybrid search, reranking, embeddings, OpenAI API, Anthropic Claude, vLLM, Ollama, LangChain, Hugging Face Transformers, LoRA/QLoRA fine-tuning, quantized models, coding agents

Machine learning: PyTorch, TensorFlow/Keras, scikit-learn, ONNX Runtime, computer vision (YOLO object detection, image classification, OpenCV), CNN and LSTM for time series

Backend: Python (FastAPI, asyncio), C#/.NET, SQL, PostgreSQL/pgvector, MongoDB, REST APIs, JavaScript (React, Node.js)

Infrastructure: Docker, Kubernetes (K3s), Linux, GitLab CI/CD, Git | **Other languages:** C++, Java, C, MATLAB

CERTIFICATIONS AND LANGUAGES

Oracle (2025–2026): Agentic AI Foundations Associate, AI Vector Search Certified Professional, OCI Generative AI Professional, OCI AI Foundations Associate | **IBM (2026):** Containers, Kubernetes and OpenShift; Docker Essentials

Languages: Italian (native), English (C1), French (A2)

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